KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5063A | 5 Credits: 4 | Program Elective | Source-grounded

Non-wovens & Technical Textiles

 To familiarize the students with the manufacturing of Non-woven fabrics and the applications areas of technical textiles

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5063A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5063A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Textile Technology
Course code	5063A
Course title	Non-wovens & Technical Textiles
Semester	5 Credits: 4
Course category	Program Elective

Item	Official information
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

10 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ￿ To familiarize the students with the manufacturing of Non-woven fabrics and the applications areas of technical textiles

Course outcomes

CO	Official outcome	Study level
CO1	14 Applying to implement the web forming methods. Execute the different bonding and finishing	See official syllabus
CO2	14 Applying techniques of non-wovens. Summarise the concept and various sectors of	See official syllabus
CO3	technical textiles- Applications of Geo-textiles 15 Applying and Agro-textiles. Application of technical textile in Medical and	See official syllabus
CO4	15 Applying Industrial sectors	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	forming methods.	2	As specified
Module 2	Execute the different bonding and finishing techniques of non-wovens	3	As specified
Module 3	Applications of Geo-textiles and Agro-textiles.	3	As specified
Module 4	Application of technical textile in Medical and Industrial sectors	2	As specified

MODULE 1

forming methods.

This module is presented from the official syllabus scope for Non-wovens & Technical Textiles. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: forming methods.
- Official duration: As specified
- Official topic entries captured: 2

- The full source excerpt is preserved below for coverage checking.

wovens and fibres used for non-woven 7 Understanding production To implement various web forming methods-

wovens and fibres used for non-woven 7 Understanding production To implement various web forming methods- is an official topic in Module 1 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify wovens and fibres used for non-woven 7 Understanding production To implement various web forming methods- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, wovens and fibres used for non-woven 7 understanding production to implement various web forming methods- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What wovens and fibres used for non-woven 7 Understanding production To implement various web forming methods- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] wovens and fibres used for non-woven 7 Understanding production To implement various web forming methods- [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

7 Applying Dry-laid, wet-laid and polymer-laid Nonwoven: Introduction - definition, classification, scope and application of nonwovens - Comparison between woven and non-woven fabrics. Sequence of process for the production of nonwovens. Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card web - randowebber. Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun bonding and Melt blown process. Laying methods: Parallel laying, Cross laying and random laying methods

7 Applying Dry-laid, wet-laid and polymer-laid Nonwoven: Introduction - definition, classification, scope and application of nonwovens - Comparison between woven and non-woven fabrics. Sequence of process for the production of nonwovens. Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card web - randowebber. Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun bonding and Melt blown process. Laying methods: Parallel laying, Cross laying and random laying methods is an official topic in Module 1 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Applying Dry-laid, wet-laid and polymer-laid Nonwoven: Introduction - definition, classification, scope and application of nonwovens - Comparison between woven and non-woven fabrics. Sequence of process for the production of nonwovens. Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card web - randowebber. Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun bonding and Melt blown process. Laying methods: Parallel laying, Cross laying and random laying methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 applying dry-laid, wet-laid and polymer-laid nonwoven: introduction - definition, classification, scope and application of nonwovens - comparison between woven and non-woven fabrics. sequence of process for the production of nonwovens. web forming: fibres used and various methods of web preparation, dry-laid - card web - randowebber. wet-laid - raw materials - web formation - polymer - laid web formation - spun bonding and melt blown process. laying methods: parallel laying, cross laying and random laying methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Applying Dry-laid, wet-laid and polymer-laid Nonwoven: Introduction - definition, classification, scope and application of nonwovens - Comparison between woven and non-woven fabrics. Sequence of process for the production of nonwovens. Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card web - randowebber. Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun bonding and Melt blown process. Laying methods: Parallel laying, Cross laying and random laying methods means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 7 Applying Dry-laid, wet-laid and polymer-laid Nonwoven: Introduction - definition, classification, scope and application of nonwovens - Comparison between woven and non-woven fabrics. Sequence of process for the production of nonwovens. Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card web - randowebber. Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun bonding and Melt blown process. Laying methods: Parallel laying, Cross laying and random laying methods □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
forming methods.
             Summarize the production process of Non-
  M1.01      wovens and fibres used for non-woven               7
Understanding
             production
             To implement various web forming methods-
  M1.02                                                         7
Applying
             Dry-laid, wet-laid and polymer-laid

Contents:
Nonwoven: Introduction - definition, classification, scope and application of
nonwovens -
Comparison between woven and non-woven fabrics. Sequence of process for the
production of nonwovens.
Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card
web
- randowebber.
Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun
bonding
and Melt blown process.
Laying methods: Parallel laying, Cross laying and random laying methods
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Execute the different bonding and finishing techniques of non-wovens

This module is presented from the official syllabus scope for Non-wovens & Technical Textiles. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Execute the different bonding and finishing techniques of non-wovens
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

5 Understanding techniques To execute the process of Mechanical, Chemical and Thermal bonding

5 Understanding techniques To execute the process of Mechanical, Chemical and Thermal bonding is an official topic in Module 2 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding techniques To execute the process of Mechanical, Chemical and Thermal bonding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding techniques to execute the process of mechanical, chemical and thermal bonding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding techniques To execute the process of Mechanical, Chemical and Thermal bonding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 5 Understanding techniques To execute the process of Mechanical, Chemical and Thermal bonding □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

5 Applying nonwovensand to test and compare parameters of each type. Discuss the Finishing and limitations of

5 Applying nonwovensand to test and compare parameters of each type. Discuss the Finishing and limitations of is an official topic in Module 2 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying nonwovensand to test and compare parameters of each type. Discuss the Finishing and limitations of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying nonwovensand to test and compare parameters of each type. discuss the finishing and limitations of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Applying nonwovensand to test and compare parameters of each type. Discuss the Finishing and limitations of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 2-□□ 5 Applying nonwovensand to test and compare parameters of each type. Discuss the Finishing and limitations of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

4 Understanding nonwovens Bonding methods: mechanical, thermal, chemical/ adhesive. Mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications. Chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens. Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through air bonding. Testing the fabric quality of each bonding method. Finishing of bonded fabrics. Fusing - methods of fusing. Discuss the limitations of non-woven fabrics. Summarise the concept and various sectors of technical textiles-

4 Understanding nonwovens Bonding methods: mechanical, thermal, chemical/ adhesive. Mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications. Chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens. Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through air bonding. Testing the fabric quality of each bonding method. Finishing of bonded fabrics. Fusing - methods of fusing. Discuss the limitations of non-woven fabrics. Summarise the concept and various sectors of technical textiles- is an official topic in Module 2 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding nonwovens Bonding methods: mechanical, thermal, chemical/ adhesive. Mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications. Chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens. Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through air bonding. Testing the fabric quality of each bonding method. Finishing of bonded fabrics. Fusing - methods of fusing. Discuss the limitations of non-woven fabrics. Summarise the concept and various sectors of technical textiles- using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding nonwovens bonding methods: mechanical, thermal, chemical/ adhesive. mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications. chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens. thermal bonding nonwoven - methods of thermal bonding hot calendaring, through air bonding. testing the fabric quality of each bonding method. finishing of bonded fabrics. fusing - methods of fusing. discuss the limitations of non-woven fabrics. summarise the concept and various sectors of technical textiles- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding nonwovens Bonding methods: mechanical, thermal, chemical/ adhesive. Mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications. Chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens. Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through air bonding. Testing the fabric quality of each bonding method. Finishing of bonded fabrics. Fusing - methods of fusing. Discuss the limitations of non-woven fabrics. Summarise the concept and various sectors of technical textiles- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding nonwovens Bonding methods: mechanical, thermal, chemical/ adhesive. Mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications. Chemically bonded nonwoven -

saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens. Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through air bonding. Testing the fabric quality of each bonding method. Finishing of bonded fabrics. Fusing - methods of fusing. Discuss the limitations of non-woven fabrics. Summarise the concept and various sectors of technical textiles- □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Execute the different bonding and finishing techniques of non-wovens

             To know various Nonwoven bonding
  M2.01                                                        5
Understanding
             techniques
             To execute the process of Mechanical,
             Chemical and Thermal bonding
  M2.02                                                        5
Applying
             nonwovensand to test and compare
             parameters of each type.
             Discuss the Finishing and limitations of
  M2.03                                                        4
Understanding
             nonwovens
             Series Test- I                                    1
Contents:
Bonding methods: mechanical, thermal, chemical/ adhesive.
Mechanically bonded webs - needle punched nonwovens - working of needling loom,
stitch bonded nonwovens with and without binding threads - description,
applications.
Chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding,
powder
bonding, and application of chemical bonded nonwovens.
Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through
air
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Applications of Geo-textiles and Agro-textiles.

This module is presented from the official syllabus scope for Non-wovens & Technical Textiles. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Applications of Geo-textiles and Agro-textiles.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

5 Understanding technical textiles

5 Understanding technical textiles is an official topic in Module 3 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding technical textiles using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding technical textiles should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding technical textiles means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 5 Understanding technical textiles [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Application and analysis of Geotextiles

Application and analysis of Geotextiles is an official topic in Module 3 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Application and analysis of Geotextiles using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, application and analysis of geotextiles should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Application and analysis of Geotextiles means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Application and analysis of Geotextiles [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Use of agro-textiles 5 Applying Technical Textiles: Definition - scope - State various applications of Technical Textiles. Geo-textiles: Definition - Basic functions–essential properties - Types of geo textiles- Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment Stabilization - fibres used and fabric particulars for these applications Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications

Use of agro-textiles 5 Applying Technical Textiles: Definition - scope - State various applications of Technical Textiles. Geo-textiles: Definition - Basic functions–essential properties - Types of geo textiles- Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment Stabilization - fibres used and fabric particulars for these applications Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications is an official topic in Module 3 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Use of agro-textiles 5 Applying Technical Textiles: Definition - scope - State various applications of Technical Textiles. Geo-textiles: Definition - Basic functions–essential properties - Types of geo textiles- Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment Stabilization - fibres used and fabric particulars for these applications Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, use of agro-textiles 5 applying technical textiles: definition - scope - state various applications of technical textiles. geo-textiles: definition - basic functions–essential properties - types of geo textiles- applications of geo textiles and analysis -road construction, railway & embankment stabilization - fibres used and fabric particulars for these applications agro-textiles: definition - applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Use of agro-textiles 5 Applying Technical Textiles: Definition - scope - State various applications of Technical Textiles. Geo-textiles: Definition - Basic functions–essential properties - Types of geo textiles- Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment Stabilization - fibres used and fabric particulars for these applications Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Use of agro-textiles 5 Applying Technical Textiles: Definition - scope - State various applications of Technical Textiles. Geo-textiles: Definition - Basic functions–essential properties - Types of geo textiles- Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment Stabilization - fibres used and fabric particulars for these applications Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Applications of Geo-textiles and Agro-textiles.
             Illustrate the scope and major applications of
  M3.01                                                            5
Understanding
             technical textiles
  M3.02      Application and analysis of Geotextiles               5
Applying

  M3.03      Use of agro-textiles                                  5
Applying

Contents:
Technical Textiles: Definition - scope - State various applications of Technical
Textiles.
Geo-textiles: Definition - Basic functions—essential properties - Types of geo
textiles-
Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment
Stabilization - fibres used and fabric particulars for these applications
Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect
and bird
netting, shade fabrics - fibres used and fabric particulars for the above
applications
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Application of technical textile in Medical and Industrial sectors

This module is presented from the official syllabus scope for Non-wovens & Technical Textiles. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours

see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Application of technical textile in Medical and Industrial sectors
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

Medical application of Technical Textiles 7 Applying Application of technical textile in industry

Medical application of Technical Textiles 7 Applying Application of technical textile in industry is an official topic in Module 4 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Medical application of Technical Textiles 7 Applying Application of technical textile in industry using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, medical application of technical textiles 7 applying application of technical textile in industry should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Medical application of Technical Textiles 7 Applying Application of technical textile in industry means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Medical application of Technical Textiles 7 Applying Application of technical textile in industry □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

sector Medical Textiles: Classification and fibres used - requirements. Detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads. Industrial Textiles: Definition - Hoses, Belts-Transmission belts & Conveyor belts - Strength test - materials used. Tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics. Bullet proof fabric construction and its strength analysis, Fire proof fabrics. Text /Reference: T/R Book Title/Author A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile T1 Institute R2 Prof. N. N. Banerjee Nonwovens manufacture, 1979 R3 N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall T. Karthik, Prabha Karan C& R. Rathinamoorthy- Nonwovens: Process, R4 Structure, Properties and Applications- Textile institute(Wood head publishing India) R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, R5 Vol.2, 1970 Textile institute R6 V Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow R7 K P Chellaman i - Yarns& Technical Textiles SITRA Publication Prof. P A Khatwani& S SYardi - NCUTE Pilot Programme-Technical textiles R8 NCUTE, IIT, New Delhi Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/

sector Medical Textiles: Classification and fibres used - requirements. Detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads. Industrial Textiles: Definition - Hoses, Belts-Transmission belts & Conveyor belts - Strength test - materials used. Tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics. Bullet proof fabric construction and its strength analysis, Fire proof fabrics. Text /Reference: T/R Book Title/Author A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile T1 Institute R2 Prof. N. N. Banerjee Nonwovens manufacture, 1979 R3 N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall T. Karthik, Prabha Karan C& R. Rathinamoorthy- Nonwovens: Process, R4 Structure, Properties and Applications- Textile institute(Wood head publishing India) R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, R5 Vol.2, 1970 Textile institute R6 V Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow R7 K P Chellaman i - Yarns& Technical Textiles SITRA Publication Prof. P A Khatwani& S SYardi - NCUTE Pilot Programme- Technical textiles R8 NCUTE, IIT, New Delhi Delhi Online resources:

Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ is an official topic in Module 4 of Non-wovens & Technical Textiles. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify sector Medical Textiles: Classification and fibres used - requirements. Detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads. Industrial Textiles: Definition - Hoses, Belts- Transmission belts & Conveyor belts - Strength test - materials used. Tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics. Bullet proof fabric construction and its strength analysis, Fire proof fabrics. Text /Reference: T/R Book Title/Author A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile T1 Institute R2 Prof. N. N. Banerjee Nonwovens manufacture, 1979 R3 N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall T. Karthik, Prabha Karan C& R. Rathinamoorthy- Nonwovens: Process, R4 Structure, Properties and Applications- Textile institute(Wood head publishing India) R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, R5 Vol.2, 1970 Textile institute R6 V Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow R7 K P Chellaman i - Yarns& Technical Textiles SITRA Publication Prof. P A Khatwani& S SYardi - NCUTE Pilot Programme- Technical textiles R8 NCUTE, IIT, New Delhi Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sector medical textiles: classification and fibres used - requirements. detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads. industrial textiles: definition - hoses, belts- transmission belts & conveyor belts - strength test - materials used. tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics. bullet proof fabric construction and its strength analysis, fire proof fabrics. text /reference: t/r book title/author a.r.horrocks&s.c.anand - hand book of technical textiles - the textile t1 institute r2 prof. n. n. banerjee nonwovens manufacture, 1979 r3 n m w john - geo textiles - blackie published in usa by chapman and hall t. karthik, prabha karan c& r. rathinamoorthy- nonwovens: process, r4 structure, properties and applications- textile institute(wood head publishing india) r s goy & j a jenkins - industrial application of textiles, textile progress, r5 vol.2, 1970 textile institute r6 v usenko - processing of man made fibers, 1979 - mir publishers moscow r7 k p chellaman i - yarns& technical textiles sitra publication prof. p a khatwani& s syardi - ncute pilot programme- technical textiles r8 ncute, iit, new delhi delhi online resources: sl.no website link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 october 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What sector Medical Textiles: Classification and fibres used - requirements. Detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads. Industrial Textiles: Definition - Hoses, Belts- Transmission belts & Conveyor belts - Strength test - materials used. Tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics. Bullet proof fabric construction and its strength analysis, Fire proof fabrics. Text /Reference: T/R Book Title/Author A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile T1 Institute R2 Prof. N. N. Banerjee Nonwovens manufacture, 1979 R3 N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall T. Karthik, Prabha Karan C& R. Rathinamoorthy- Nonwovens: Process, R4 Structure, Properties and Applications- Textile institute(Wood head publishing India) R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, R5 Vol.2, 1970 Textile institute R6 V Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow R7 K P Chellaman i - Yarns& Technical Textiles SITRA Publication Prof. P A Khatwani& S SYardi - NCUTE Pilot Programme- Technical textiles R8 NCUTE, IIT, New Delhi Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ sector Medical Textiles: Classification and fibres used - requirements. Detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads. Industrial Textiles: Definition - Hoses, Belts- Transmission belts & Conveyor belts - Strength test - materials used. Tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics. Bullet proof fabric construction and its strength analysis, Fire proof fabrics. Text /Reference: T/R Book Title/Author A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile T1 Institute R2 Prof. N. N. Banerjee Nonwovens manufacture, 1979 R3 N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall T. Karthik, Prabha Karan C& R. Rathinamoorthy- Nonwovens: Process, R4 Structure, Properties and Applications- Textile institute(Wood head publishing India) R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, R5 Vol.2, 1970 Textile institute R6 V Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow R7 K P Chellaman i - Yarns& Technical Textiles SITRA Publication Prof. P A Khatwani& S SYardi - NCUTE Pilot Programme- Technical textiles R8 NCUTE, IIT, New Delhi Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Application of technical textile in Medical and Industrial sectors

  M4.01      Medical application of Technical Textiles            7
Applying
             Application of technical textile in industry
                                                                  8
Applying
  M4.02      sector
             Series Test - II                                     1

Contents:
Medical Textiles: Classification and fibres used - requirements.
Detailed study and application of technical textiles in: implantable - non-
implantable -
extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns,
masks,
sanitary pads.
Industrial Textiles: Definition - Hoses, Belts- Transmission belts & Conveyor
belts -
Strength test - materials used. Tyre cords - flow chart for tyre cord
manufacturing -
physical and mechanical requirements of tyre fabrics.
Bullet proof fabric construction and its strength analysis, Fire proof fabrics.
Text /Reference:

  T/R                                    Book Title/Author
          A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain wovens and fibres used for non-woven 7 Understanding production To implement various web forming methods-. (3 marks)

Answer: Begin with the standard definition or identity of *wovens and fibres used for non-woven 7 Understanding production To implement various web forming methods-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 7 Applying Dry-laid, wet-laid and polymer-laid Nonwoven: Introduction - definition, classification, scope and application of nonwovens - Comparison between woven and non-woven fabrics. Sequence of process for the production of nonwovens. Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card web - randowebber. Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun bonding and Melt blown process. Laying methods: Parallel laying, Cross laying and random laying methods. (3 marks)

Answer: Begin with the standard definition or identity of *7 Applying Dry-laid, wet-laid and polymer-laid Nonwoven: Introduction - definition, classification, scope and application of nonwovens - Comparison between woven and non-woven fabrics. Sequence of process for the production of nonwovens. Web forming: Fibres used and Various methods of web preparation, Dry-laid - Card web - randowebber. Wet-laid - raw materials - web formation - Polymer - laid web formation - Spun bonding and Melt blown process. Laying methods: Parallel laying, Cross laying and random laying methods*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 5 Understanding techniques To execute the process of Mechanical, Chemical and Thermal bonding. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding techniques To execute the process of Mechanical, Chemical and Thermal bonding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Applying nonwovensand to test and compare parameters of each type. Discuss the Finishing and limitations of. (3 marks)

Answer: Begin with the standard definition or identity of *5 Applying nonwovensand to test and compare parameters of each type. Discuss the Finishing and limitations of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding nonwovens Bonding methods: mechanical, thermal, chemical/ adhesive. Mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications. Chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens. Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through air bonding. Testing the fabric quality of each bonding method. Finishing of bonded fabrics. Fusing - methods of fusing. Discuss the limitations of non-woven fabrics. Summarise the concept and various sectors of technical textiles-. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding nonwovens Bonding methods: mechanical, thermal, chemical/ adhesive. Mechanically bonded webs - needle punched nonwovens - working of needling loom, stitch bonded nonwovens with and without binding threads - description, applications. Chemically bonded nonwoven - saturation, spray adhesive bonding, print bonding, powder bonding, and application of chemical bonded nonwovens. Thermal bonding nonwoven - methods of thermal bonding Hot calendaring, Through air bonding. Testing the fabric quality of each bonding method. Finishing of bonded fabrics. Fusing - methods of fusing. Discuss the limitations of non-woven fabrics. Summarise the concept and various sectors of technical textiles-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 5 Understanding technical textiles. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding technical textiles*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Application and analysis of Geotextiles. (3 marks)

Answer: Begin with the standard definition or identity of *Application and analysis of Geotextiles*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Use of agro-textiles 5 Applying Technical Textiles: Definition - scope - State various applications of Technical Textiles. Geo-textiles: Definition - Basic functions–essential properties - Types of geo textiles- Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment Stabilization - fibres used and fabric particulars for these applications Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications. (3 marks)

Answer: Begin with the standard definition or identity of *Use of agro-textiles 5 Applying Technical Textiles: Definition - scope - State various applications of Technical Textiles. Geo-textiles: Definition - Basic functions–essential properties - Types of geo textiles-*

Applications of Geo Textiles and analysis -Road Construction, Railway & Embankment Stabilization - fibres used and fabric particulars for these applications Agro-textiles: Definition - Applications - seed bed protection, crop covers, insect and bird netting, shade fabrics - fibres used and fabric particulars for the above applications. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Module 4

Define or explain Medical application of Technical Textiles 7 Applying Application of technical textile in industry. (3 marks)

Answer: Begin with the standard definition or identity of *Medical application of Technical Textiles 7 Applying Application of technical textile in industry*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

Define or explain sector Medical Textiles: Classification and fibres used - requirements. Detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads. Industrial Textiles: Definition - Hoses, Belts- Transmission belts & Conveyor belts - Strength test - materials used. Tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics. Bullet proof fabric construction and its strength analysis, Fire proof fabrics. Text /Reference: T/R Book Title/Author A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile T1 Institute R2 Prof. N. N. Banerjee Nonwovens manufacture, 1979 R3 N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall T. Karthik, Prabha Karan C& R. Rathinamoorthy- Nonwovens: Process, R4 Structure, Properties and Applications- Textile institute(Wood head publishing India) R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, R5 Vol.2, 1970 Textile institute R6 V

Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow R7 K P Chellaman i - Yarns& Technical Textiles SITRA Publication Prof. P A Khatwani& S SYardi - NCUTE Pilot Programme- Technical textiles R8 NCUTE, IIT, New Delhi Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/. (3 marks)

Answer: Begin with the standard definition or identity of *sector Medical Textiles: Classification and fibres used - requirements. Detailed study and application of technical textiles in: implantable - non-implantable - extracorporeal devices and Healthcare / hygiene products - theatre drapes, gowns, masks, sanitary pads. Industrial Textiles: Definition - Hoses, Belts- Transmission belts & Conveyor belts - Strength test - materials used. Tyre cords - flow chart for tyre cord manufacturing - physical and mechanical requirements of tyre fabrics. Bullet proof fabric construction and its strength analysis, Fire proof fabrics. Text /Reference: T/R Book Title/Author A.R.Horrocks&S.C.Anand - Hand book of Technical Textiles - The Textile T1 Institute R2 Prof. N. N. Banerjee Nonwovens manufacture, 1979 R3 N M W John - Geo Textiles - Blackie Published in USA by Chapman and Hall T. Karthik, Prabha Karan C& R. Rathinamoorthy- Nonwovens: Process, R4 Structure, Properties and Applications- Textile institute(Wood head publishing India) R S Goy & J A Jenkins - Industrial application of textiles, Textile progress, R5 Vol.2, 1970 Textile institute R6 V Usenko - Processing of Man Made fibers, 1979 - MIR Publishers Moscow R7 K P Chellaman i - Yarns& Technical Textiles SITRA Publication Prof. P A Khatwani& S SYardi - NCUTE Pilot Programme- Technical textiles R8 NCUTE, IIT, New Delhi Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **wovens and fibres used for non-woven 7 Understanding production To implement various web forming methods-**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **5 Understanding techniques To execute the process of Mechanical, Chemical and Thermal bonding**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Understanding technical textiles**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Medical application of Technical Textiles 7 Applying Application of technical textile in industry**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- according/ (accessed on 20 October 2013)
- https://nptel.ac.in/courses/116/102/116102026/
- https://nptel.ac.in/courses/116/102/116102016/ https://swayam.gov.in/ http://eacharya.inflibnet.ac.in/vidya-mitra/

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